

Smartphone Data Analysis

Presented By: Tahnim Mahir





About Smartphone

The smartphone dataset contains 981 samples with features like battery life, camera quality, and processor speed etc.

It categorizes smartphones into low, medium, and high price ranges, aiding in market trend analysis and consumer preference insights.

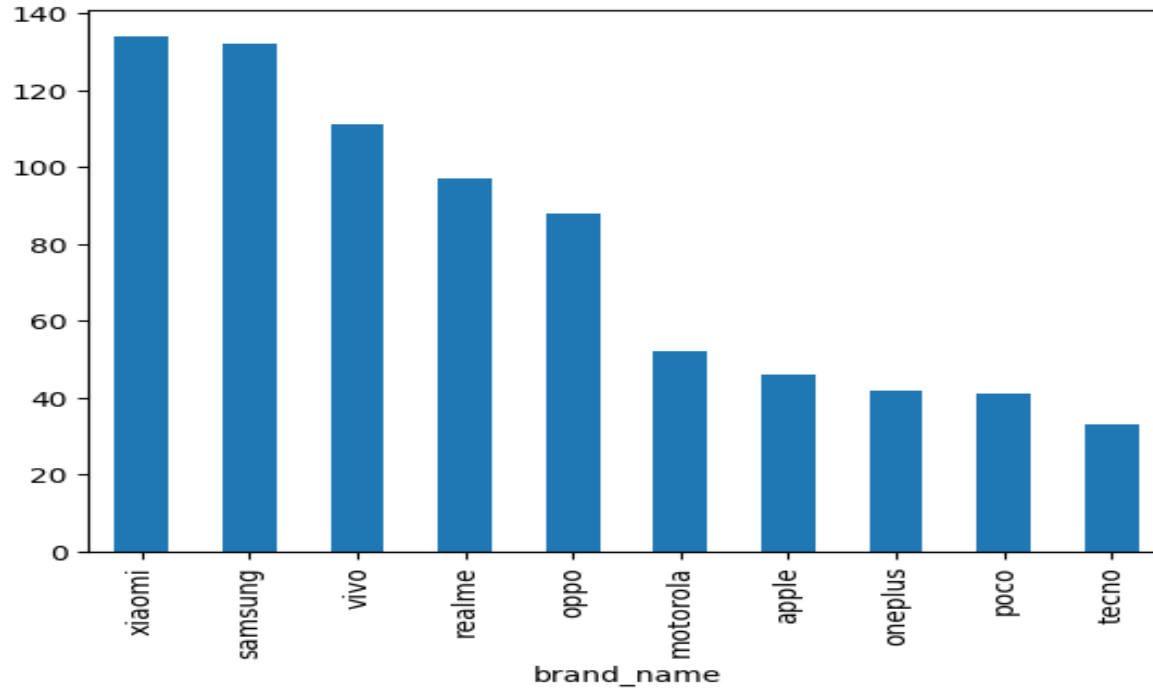
Key Findings

Based on your analysis, summarize the key findings. Here's an example format:

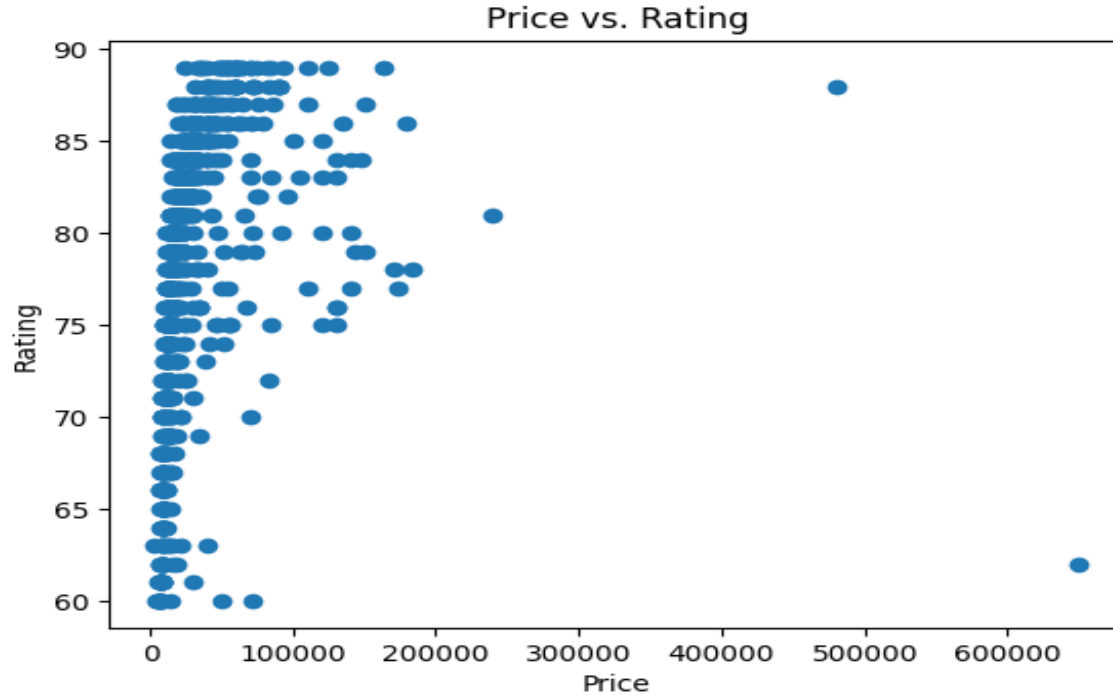
- **Price and Processor Speed:** Higher processor speeds tend to be associated with higher prices.
- **Brand Influence:** Certain brands, such as Apple, Samsung, have a higher average price.
- **Feature Influence:** Phones with 5G and NFC features tend to be priced higher.
- **Rating Factors:** Higher-rated phones often have better specifications, such as more cores or higher screen refresh rates.



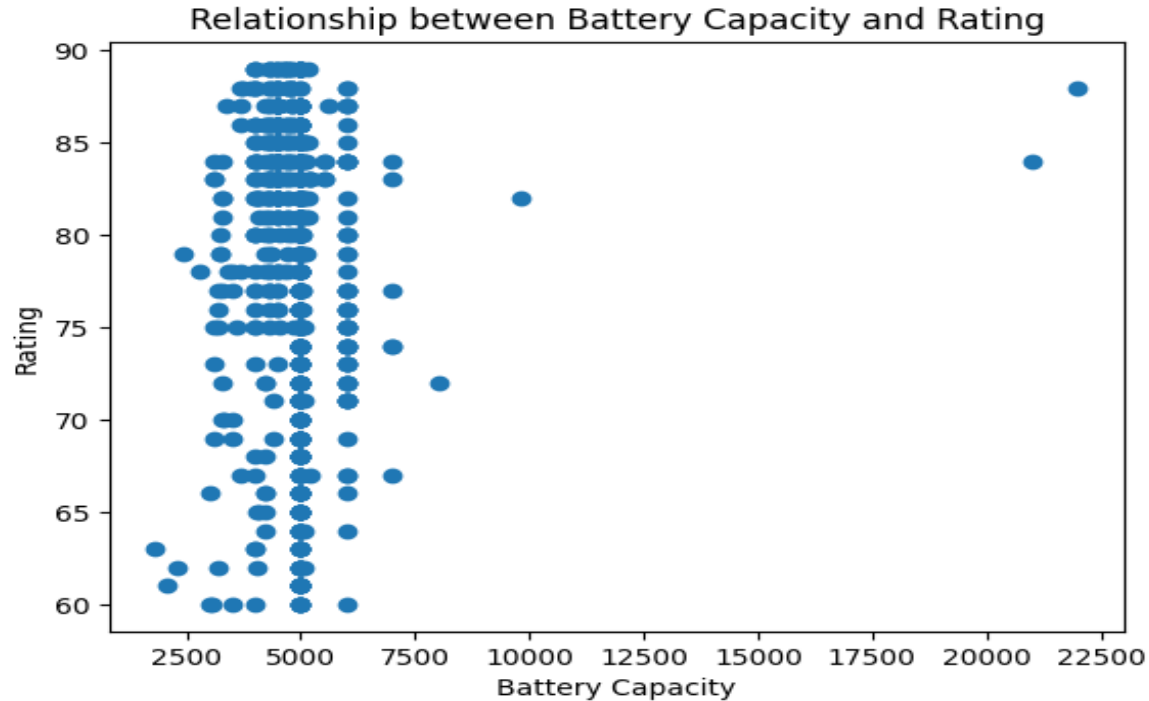
Smartphone Brands



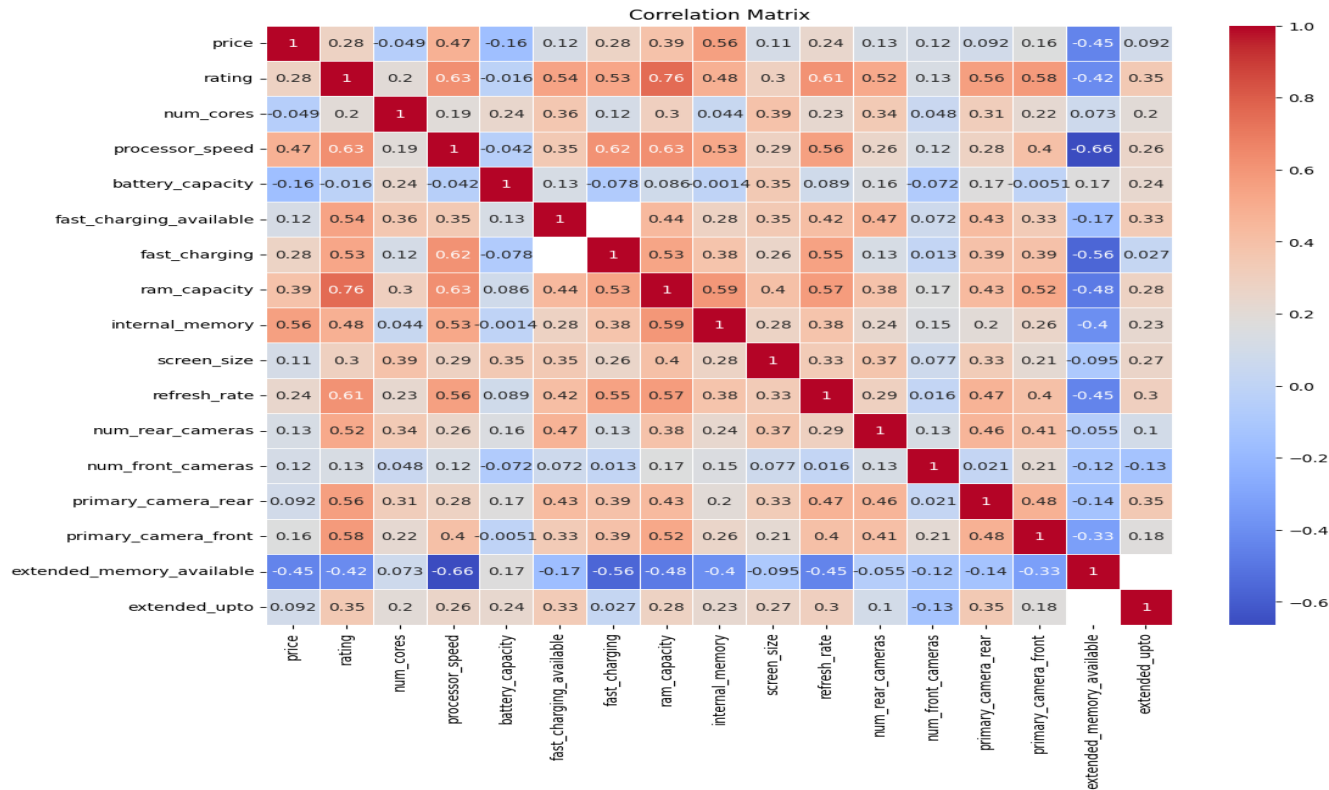
Price vs Rating



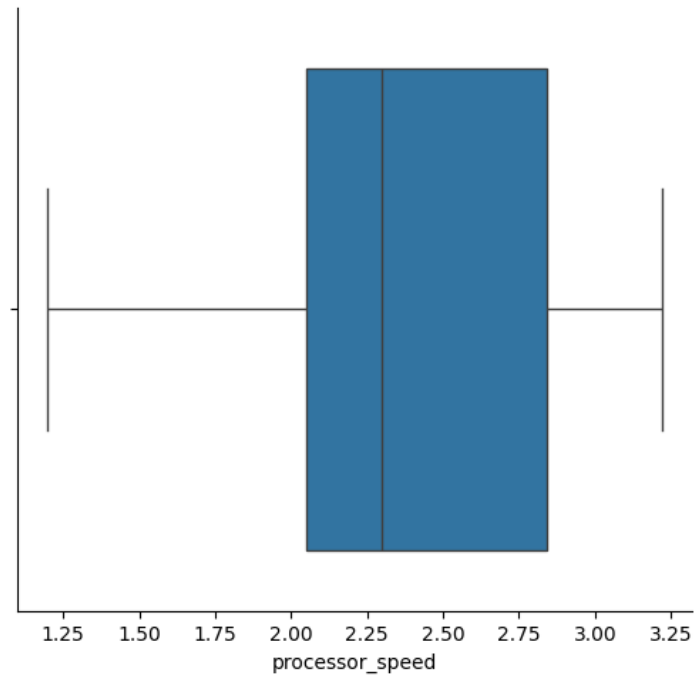
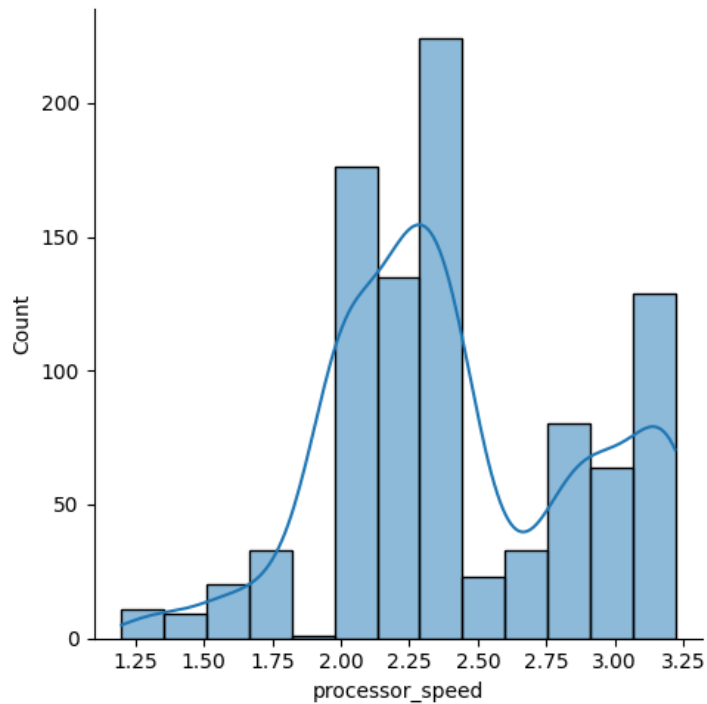
Battery capacity vs Rating



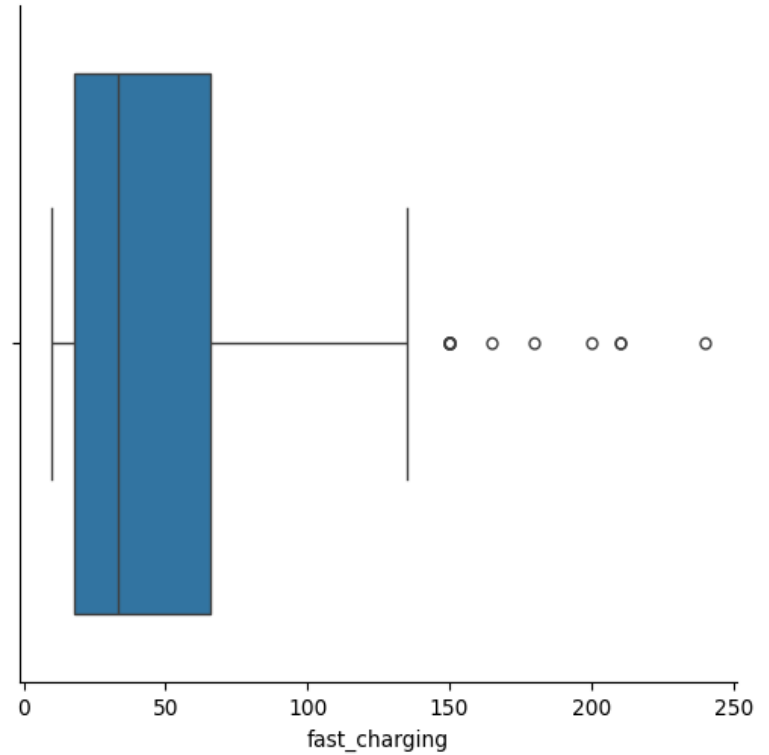
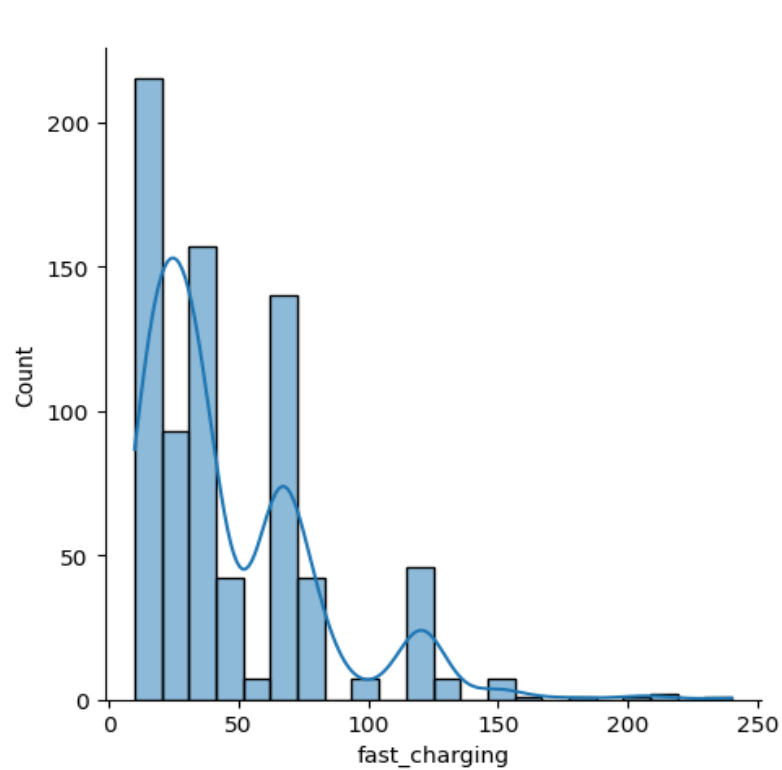
Correlation Matrix



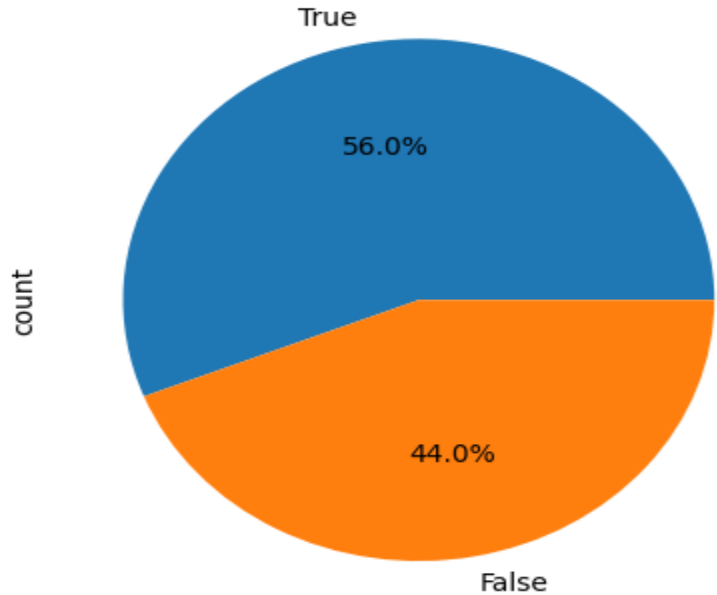
Processor Speed



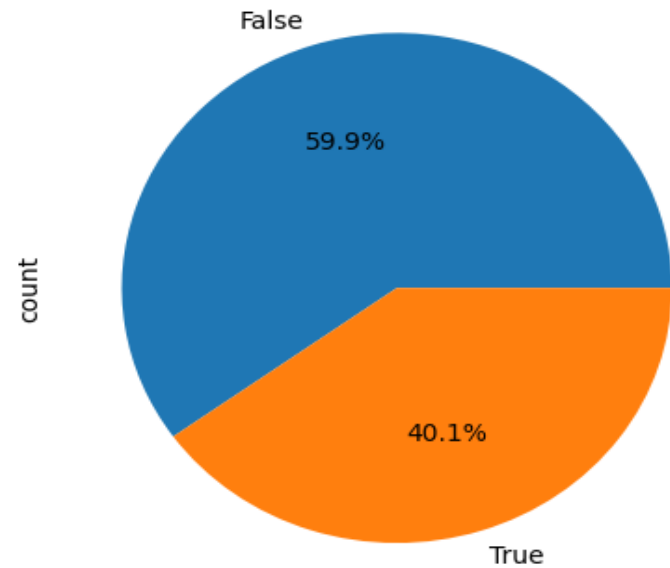
Fast Charging



5G and NFC Support



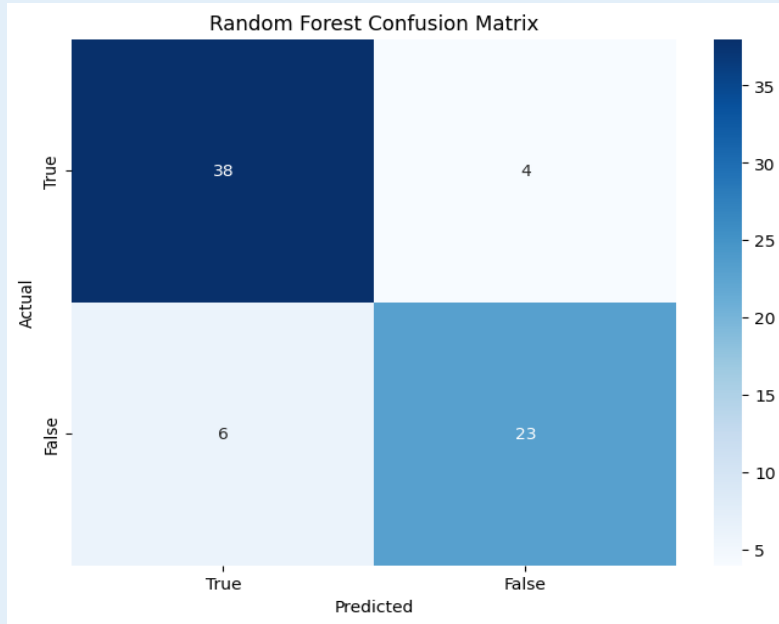
5G Support



NFC Support

Machine Learning

Model Evaluation



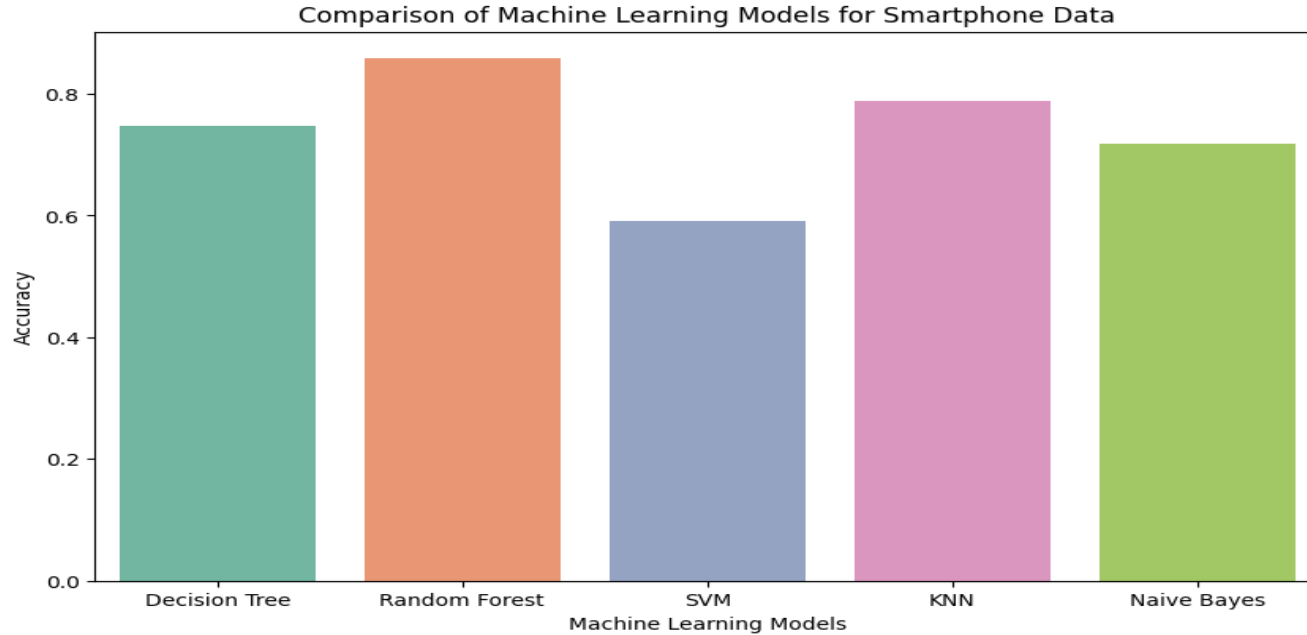
Random Forest Classifier
Accuracy: 0.8591549295774648

Classification Report:

	precision	recall	f1-score	support
False	0.86	0.90	0.88	42
True	0.85	0.79	0.82	29
accuracy			0.86	71
macro avg	0.86	0.85	0.85	71
weighted avg	0.86	0.86	0.86	71

We Try 5 different ML Algorithm. So best result we find using Random Forest Algorithm

Model Comparison

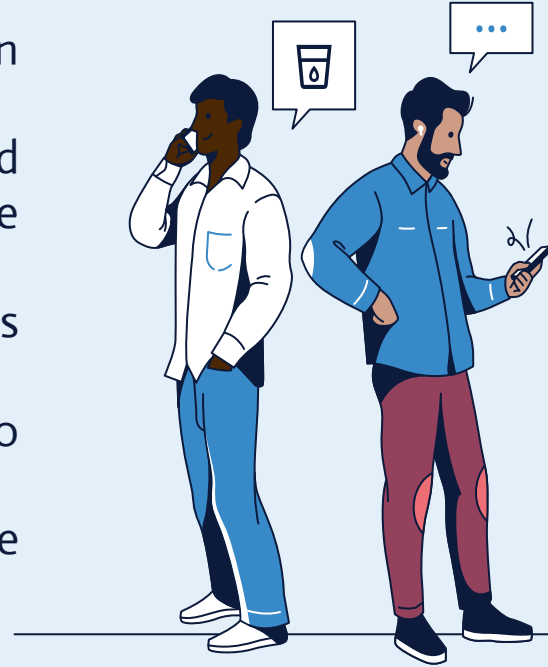


Business Simulation

- **Manufacturers:** Focus on optimizing battery life to improve user satisfaction and increase sales. And optimize internal memory configurations for different price ranges.
- **Retailers:** Highlight battery capacity as a key feature when marketing high-rated smartphones.
- **App Developers:** Optimize apps for better battery efficiency to enhance user experience and boost app ratings.
- **Correlation Insight:** Higher-priced smartphones tend to have more internal storage.

Recommendation

- Optimize battery life to boost user satisfaction and sales.
- Analyze top-rated brands to understand consumer preferences and develop competitive products.
- Improve smartphone quality by examining ratings for different processor brands.
- Position smartphones by screen size and price to meet diverse customer needs and budgets.
- Optimize apps for battery efficiency to enhance user experience and app ratings.



Conclusion

By leveraging the insights gained from the smartphone data analysis, manufacturers, retailers, and app developers can make informed decisions to improve product quality, enhance user experience, and ultimately drive business success.

We gain Mean Squared Error is 72583974.88 for using linear regression model. And Highest accuracy find using Random Forest Algorithm is 85%

Thank You